

LAPORAN PRAKTIKUM

Di butuhkan untuk memenuhi kebutuhan tugas mata kuliah system operasi

Dosen : Triawan adi cahyanto M.kom



Oleh

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```

naran@Naran:~$ free -h
              total        used        free      shared  buff/cache   avail
Mem:           3.7Gi        430Mi        3.3Gi         3.4Mi        132Mi        3
Swap:           1.0Gi           0B         1.0Gi
naran@Naran:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      2
On-line CPU(s) list:         0,1
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Celeron(R) N4020 CPU @ 1.10GHz
CPU family:                  6
Model:                      122
Thread(s) per core:         1
Core(s) per socket:         2
Socket(s):                   1
Stepping:                    8
BogoMIPS:                    2188.79
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr p
ge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss
ht syscall nx pdpe1gb rdtscp lm constant_tsc arch
_perfmom rep_good nopl xtopology tsc_reliable nons
top_tsc cpuid tsc_known_freq pni pclmulqdq vmx sss
e3 cx16 pdcm sse4_1 sse4_2 x2apic movbe popcnt tsc
_deadline_timer aes xsave rdrand hypervisor lahf_l
m 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced
tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust sm
ep erms rdseed smap clflushopt sha_ni xsaveopt xsa
vec xgetbv1 xsaves vnmi umip rdpid md_clear arch_c
apabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       48 KiB (2 instances)
  L1i:                       64 KiB (2 instances)

```

```

C:\Users\naran>wsl
naran@Naran:/mnt/c/Users/naran$ cd
naran@Naran:~$ uname -a
Linux Naran 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun
 5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
naran@Naran:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:       24.04
Codename:      noble
naran@Naran:~$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  0.3  21648 12656 ?        Ss   18:41   0:02 /sbin/init
root         2  0.0  0.0   3072  1920 ?        Sl   18:41   0:00 /init
root         7  0.0  0.0   3072  1792 ?        Sl   18:41   0:00 plan9 --c
ontrol-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        46  0.0  0.3  66820 15644 ?        S<s  18:41   0:01 /usr/lib/
systemd/systemd-journald
root       102  0.0  0.1  25136  6272 ?        Ss   18:42   0:00 /usr/lib/
systemd/systemd-udevd
systemd+   111  0.0  0.3  21456 12672 ?        Ss   18:42   0:00 /usr/lib/
systemd/systemd-resolved
systemd+   112  0.0  0.1  91024  7808 ?        Ssl  18:42   0:00 /usr/lib/
systemd/systemd-timesyncd
root       155  0.0  0.0   4236  2688 ?        Ss   18:42   0:00 /usr/sbin
/cron -f -p
message+   156  0.0  0.1   9588  5120 ?        Ss   18:42   0:00 @dbus-dae
mon --system --address=systemd: --nofork --nopidfile --systemd-activation --
syslog-only
root       172  0.0  0.2  17968  8448 ?        Ss   18:42   0:00 /usr/lib/
systemd/systemd-logind
root       174  0.0  0.3 1681852 12544 ?        Ssl  18:42   0:00 /usr/libe
xec/wsl-pro-service -vv
root       177  0.0  0.0   3160  1792 hvc0     Ss+  18:42   0:00 /sbin/age
tty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root       189  0.0  0.0   3116  1920 tty1      Ss+  18:42   0:00 /sbin/age

```

```

naran@Naran:~/praktikum1$ nano fcfs.c
naran@Naran:~/praktikum1$ gcc -o fcfs fcfs.c
./fcfs
Enter total processes:
1
Enter burst time for each process:
P1: nano sjf.c

Process BT      WT      TAT
P1      545522177      0      545522177

Average Waiting Time: 0.00
Average Turnaround Time: 545522176.00
naran@Naran:~/praktikum1$ nano sjf.c
naran@Naran:~/praktikum1$ nano rr.c

```

```

naran@Naran:~/praktikum1$ swapon --show
NAME      TYPE      SIZE USED PRIO
/dev/sdc partition 1G  0B  -2
naran@Naran:~/praktikum1$ ps aux --sort=-%mem | head -5
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root      209  0.0  0.5 107032 22400 ?        Ssl  17:54   0:00 /usr/bin/
python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-fo
r-signal
root       52  0.0  0.4  66752 15688 ?        S<s  17:54   0:03 /usr/lib/
systemd/systemd-journald
root      174  0.0  0.3 1681852 12800 ?        Ssl  17:54   0:00 /usr/libe
xec/wsl-pro-service -vv
root       1  0.1  0.3  21788 12776 ?        Ss   17:54   0:03 /sbin/ini
t
naran@Naran:~/praktikum1$ nano memory_sim.c
naran@Naran:~/praktikum1$ gcc -o memory_sim memory_sim.c
./memory_sim
=== Memory Allocation Simulation ===
Allocated 4 MB successfully
Memory freed at step 1
Allocated 8 MB successfully
Memory freed at step 2
Allocated 12 MB successfully
Memory freed at step 3
Allocated 16 MB successfully
Memory freed at step 4
Allocated 20 MB successfully
Memory freed at step 5
Simulation done.

```

```

naran@Naran:~/praktikum1$ cat /proc/uptime
3152.82 6151.08
naran@Naran:~/praktikum1$ free -h
              total        used        free      shared  buff/cache   avail
Mem:           3.7Gi         418Mi         3.3Gi         3.4Mi         102Mi         3
Swap:           1.0Gi           0B           1.0Gi
naran@Naran:~/praktikum1$ cat /proc/meminfo | head -15
MemTotal:        3919608 kB
MemFree:         3507216 kB
MemAvailable:    3490380 kB
Buffers:          836 kB
Cached:          86828 kB
SwapCached:        0 kB
Active:          37204 kB
Inactive:        122420 kB
Active(anon):     2448 kB
Inactive(anon):   73032 kB
Active(file):     34756 kB
Inactive(file):   49388 kB
Unevictable:        0 kB
Mlocked:          0 kB
SwapTotal:       1048576 kB
naran@Naran:~/praktikum1$ swapon --show
NAME      TYPE      SIZE USED PRIO
/dev/sdc partition 1G  0B  -2
naran@Naran:~/praktikum1$ ps aux --sort=-%mem | head -5
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root       209  0.0  0.5 107032 22400 ?        Ssl  17:54   0:00 /usr/bin/
python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root        52  0.0  0.4  66752 15688 ?        S<s  17:54   0:03 /usr/lib/
systemd/systemd-journald
root       174  0.0  0.3 1681852 12800 ?        Ssl  17:54   0:00 /usr/libe
xec/wsl-pro-service -vv
root        1  0.1  0.3  21788 12776 ?        Ss   17:54   0:03 /sbin/init

```

```

naran@Naran:~/praktikum1$ cat /proc/meminfo | head -15
MemTotal:        3919608 kB
MemFree:         3517692 kB
MemAvailable:    3500212 kB
Buffers:          564 kB
Cached:          85804 kB
SwapCached:        0 kB
Active:          36892 kB
Inactive:        118136 kB
Active(anon):     2484 kB
Inactive(anon):   69688 kB
Active(file):     34408 kB
Inactive(file):   48448 kB
Unevictable:        0 kB
Mlocked:          0 kB
SwapTotal:       1048576 kB
naran@Naran:~/praktikum1$ cat /proc/self/status
Name:   cat
Umask:  0022
State:  R (running)
Tgid:   1109
Ngid:   0
Pid:    1109
PPid:   295
TracerPid: 0
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
FDSize: 256
Groups: 4 24 27 30 46 100 1000
NStgid: 1109
NSpid:  1109
NSpgid: 1109
NSSid:  295
Kthreid: 0
VmPeak: 3268 kB
VmSize: 3268 kB
VmLck:  0 kB
VmPin:  0 kB
VmHWM:  1536 kB
VmRSS:  1536 kB
RssAnon: 0 kB

```

```
naran@Naran:~/praktikum1$ ls /proc | head
1
105
1099
1100
135
141
161
162
172
174
naran@Naran:~/praktikum1$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
```

```
insmod: ERROR: could not load module hello_module.ko: No such file or directory
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=2 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000]   Intel GenuineIntel
[ 0.000000]   AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x0000000000f7ffffff] usable
[ 0.000000] BIOS-e820: [mem 0x000000000100000000-0x0000000001025fffff] usable
```

```
naran@Naran:~/praktikum1$ objdump -d os_specific | head -50
os_specific:          file format elf64-x86-64

Disassembly of section .init:

0000000000001000 <_init>:
1000:      f3 0f 1e fa                endbr64
1004:      48 83 ec 08                sub    $0x8,%rsp
1008:      48 8b 05 d9 2f 00 00      mov    0x2fd9(%rip),%rax        # 3fa0 <GLOBAL_OFFSET_TABLE+0x8>
e8 <__gmon_start__@Base>
100f:      48 85 c0                    test   %rax,%rax
1012:      74 02                      je     1016 <_init+0x16>
1014:      ff d0                      call   *%rax
1016:      48 83 c4 08                add    $0x8,%rsp
101a:      c3                          ret

Disassembly of section .plt:

0000000000001020 <.plt>:
1020:      ff 35 7a 2f 00 00          push   0x2f7a(%rip)             # 3fa0 <GLOBAL_OFFSET_TABLE+0x8>
1026:      ff 25 7c 2f 00 00          jmp    *0x2f7c(%rip)           # 3fa8 <GLOBAL_OFFSET_TABLE+0x10>
102c:      0f 1f 40 00                nopl   0x0(%rax)
1030:      f3 0f 1e fa                endbr64
1034:      68 00 00 00 00            push   $0x0
1039:      e9 e2 ff ff ff          jmp     1020 <_init+0x20>
103e:      66 90                    xchg   %ax,%ax
1040:      f3 0f 1e fa                endbr64
1044:      68 01 00 00 00            push   $0x1
1049:      e9 d2 ff ff ff          jmp     1020 <_init+0x20>
104e:      66 90                    xchg   %ax,%ax
1050:      f3 0f 1e fa                endbr64
1054:      68 02 00 00 00            push   $0x2
1059:      e9 c2 ff ff ff          jmp     1020 <_init+0x20>
105e:      66 90                    xchg   %ax,%ax
```

```

[ 9] .gnu.version_r  VERNEED      0000000000005b8  000005b8
0000000000000040 0000000000000000 A      7      1      8
[10] .rela.dyn        RELA        00000000000005f8  000005f8
00000000000000c0 0000000000000018 A      6      0      8
[11] .rela.plt        RELA        00000000000006b8  000006b8
0000000000000078 0000000000000018 AI     6     24     8
[12] .init            PROGBITS     0000000000001000  00001000
000000000000001b 0000000000000000 AX      0      0      4
[13] .plt             PROGBITS     0000000000001020  00001020
0000000000000060 0000000000000010 AX      0      0     16
[14] .plt.got         PROGBITS     0000000000001080  00001080
0000000000000010 0000000000000010 AX      0      0     16
[15] .plt.sec         PROGBITS     0000000000001090  00001090
0000000000000050 0000000000000010 AX      0      0     16
[16] .text            PROGBITS     00000000000010e0  000010e0
00000000000001e5 0000000000000000 AX      0      0     16
[17] .fini            PROGBITS     00000000000012c8  000012c8
000000000000000d 0000000000000000 AX      0      0      4
[18] .rodata          PROGBITS     0000000000002000  00002000
000000000000007a 0000000000000000 A      0      0      8
[19] .eh_frame_hdr    PROGBITS     000000000000207c  0000207c
000000000000003c 0000000000000000 A      0      0      4
[20] .eh_frame        PROGBITS     00000000000020b8  000020b8
00000000000000cc 0000000000000000 A      0      0      8
[21] .init_array       INIT_ARRAY    0000000000003d98  00002d98
0000000000000008 0000000000000008 WA      0      0      8
[22] .fini_array       FINI_ARRAY    0000000000003da0  00002da0
0000000000000008 0000000000000008 WA      0      0      8
[23] .dynamic          DYNAMIC      0000000000003da8  00002da8
00000000000001f0 0000000000000010 WA      7      0      8
[24] .got              PROGBITS     0000000000003f98  00002f98
0000000000000068 0000000000000008 WA      0      0      8

```

```

naran@Naran:~/praktikum1$ file os_spesicific
os_spesicific: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dyna
mically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=6bef0
dea46d26ca019b34bfca4d0decf84943a56, for GNU/Linux 3.2.0, not stripped
naran@Naran:~/praktikum1$ readelf -h os_spesicific
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:             ELF64
  Data:              2's complement, little endian
  Version:           1 (current)
  OS/ABI:             UNIX - System V
  ABI Version:       0
  Type:              DYN (Position-Independent Executable file)
  Machine:           Advanced Micro Devices X86-64
  Version:           0x1
  Entry point address: 0x10e0
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14200 (bytes into file)
  Flags:             0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30

```

```

naran@Naran:~/praktikum1$ ldd program
        linux-vdso.so.1 (0x00007ffdce963000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007071ab000000)
        /lib64/ld-linux-x86-64.so.2 (0x00007071ab264000)
naran@Naran:~/praktikum1$ gcc main.c math_ops.c -o program_static -static
naran@Naran:~/praktikum1$ gcc main.c math_ops.c -o program_dynamic
naran@Naran:~/praktikum1$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 naran naran 16K Oct 15 18:22 program_dynamic
-rwxr-xr-x 1 naran naran 767K Oct 15 18:22 program_static
naran@Naran:~/praktikum1$ gcc -c -fPIC math_ops.c -o math_ops.o
naran@Naran:~/praktikum1$ gcc main.c -L. -lmathops -o program_shared
/usr/bin/ld: cannot find -lmathops: No such file or directory
collect2: error: ld returned 1 exit status
naran@Naran:~/praktikum1$ LD_LIBRARY_PATH=. ./program_shared
-bash: ./program_shared: No such file or directory
naran@Naran:~/praktikum1$ nano os_spesific.c
naran@Naran:~/praktikum1$ gcc -o os_specific os_specific.c
cc1: fatal error: os_specific.c: No such file or directory
compilation terminated.
naran@Naran:~/praktikum1$ gcc -o os_specisific os_specific.c
cc1: fatal error: os_specific.c: No such file or directory
compilation terminated.
naran@Naran:~/praktikum1$ gcc -o os_spesific os_specific.c
cc1: fatal error: os_specific.c: No such file or directory
compilation terminated.
naran@Naran:~/praktikum1$ gcc -o os_spesific os_spesific.c
naran@Naran:~/praktikum1$ ./os_spesific
=== OS-Specific Application ===
OS: Linux
System: Linux
Release: 6.6.87.2-microsoft-standard-WSL2
Version: #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
Linux-specific: Process ID = 592

```

```

naran@Naran:~/praktikum1$ gcc main.c math_ops.c -o program_static -static
naran@Naran:~/praktikum1$ gcc main.c math_ops.c -o program_dynamic
naran@Naran:~/praktikum1$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 naran naran 16K Oct 15 18:22 program_dynamic
-rwxr-xr-x 1 naran naran 767K Oct 15 18:22 program_static
naran@Naran:~/praktikum1$ gcc -c -fPIC math_ops.c -o math_ops.o
naran@Naran:~/praktikum1$ gcc main.c -L. -lmathops -o program_shared
/usr/bin/ld: cannot find -lmathops: No such file or directory
collect2: error: ld returned 1 exit status
naran@Naran:~/praktikum1$ LD_LIBRARY_PATH=. ./program_shared
-bash: ./program_shared: No such file or directory

```

```

naran@Naran:~/praktikum1$ ldd program
        linux-vdso.so.1 (0x00007ffdce963000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007071ab000000)
        /lib64/ld-linux-x86-64.so.2 (0x00007071ab264000)
naran@Naran:~/praktikum1$ |

```



```

naran@Naran:~/praktikum1$ nm main.o
                 U add
0000000000000000 T main
                 U multiply
                 U printf
naran@Naran:~/praktikum1$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
naran@Naran:~/praktikum1$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
                 w _ITM_deregisterTMCloneTable
                 w _ITM_registerTMCloneTable
0000000000002148 r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
000000000000038c r __abi_tag
0000000000004010 B __bss_start
                 w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_dtors_aux
0000000000003dc0 d __do_global_dtors_aux_fini_array_entry
0000000000004008 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
                 w __gmon_start__
                 U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
                 U printf@GLIBC_2.2.5

```

```

C:\Users\naran>wsl
cdnaran@Naran:/mnt/c/Users/naran$ cd ~/praktikum1
naran@Naran:~/praktikum1$ nano math_ops.h
naran@Naran:~/praktikum1$ nano math_ops.c
naran@Naran:~/praktikum1$ nano main.c
naran@Naran:~/praktikum1$ gcc -E main.c -o main.i
naran@Naran:~/praktikum1$ gcc -E math_ops.c -o math_ops.i
naran@Naran:~/praktikum1$ gcc -S main.c -o main.s
naran@Naran:~/praktikum1$ gcc -S math_ops.c -o math_ops.s
naran@Naran:~/praktikum1$ gcc -c main.c -o main.o
naran@Naran:~/praktikum1$ gcc -c math_ops.c -o math_ops.o
naran@Naran:~/praktikum1$ gcc main.o math_ops.o -o program
naran@Naran:~/praktikum1$ ./program
5 + 3 = 8
5 * 3 = 15

```

```

naran@Naran:~/praktikum1$ sudo journalctl -t my_daemon -n 10
[sudo] password for naran:
Oct 15 16:58:59 Naran my_daemon[563]: Daemon is running...
Oct 15 16:59:09 Naran my_daemon[612]: Daemon is running...
Oct 15 16:59:29 Naran my_daemon[563]: Daemon is running...
Oct 15 16:59:37 Naran my_daemon[612]: Daemon is running...
Oct 15 16:59:57 Naran my_daemon[563]: Daemon is running...
Oct 15 17:00:04 Naran my_daemon[612]: Daemon is running...
Oct 15 17:00:24 Naran my_daemon[563]: Daemon is running...
Oct 15 17:00:32 Naran my_daemon[612]: Daemon is running...
Oct 15 17:00:52 Naran my_daemon[563]: Daemon is running...
Oct 15 17:00:59 Naran my_daemon[612]: Daemon is running...
naran@Naran:~/praktikum1$ sudo nano /etc/systemd/system/my_daemon.service
naran@Naran:~/praktikum1$ sudo systemctl start my_daemon
naran@Naran:~/praktikum1$ sudo systemctl daemon-reload
naran@Naran:~/praktikum1$ sudo systemctl status my_daemon
● my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; prese
   Active: active (running) since Wed 2025-10-15 17:07:13 WIB; 26s ago
     Main PID: 428 (my_daemon)
       Tasks: 1 (limit: 4581)
      Memory: 220.0K (peak: 572.0K)
         CPU: 5ms
       CGroup: /system.slice/my_daemon.service
               └─428 /home/naran/praktikum1/my_daemon

Oct 15 17:07:13 Naran systemd[1]: Starting my_daemon.service - My Custom Da
Oct 15 17:07:13 Naran my_daemon[428]: My daemon started
Oct 15 17:07:13 Naran my_daemon[428]: Daemon is running...
Oct 15 17:07:13 Naran systemd[1]: Started my_daemon.service - My Custom Dae
lines 1-14/14 (END)

```

```

cd naran@Naran:~/mnt/c/Users/naran$ cd ~/praktikum1
naran@Naran:~/praktikum1$ sudo ./my_daemon &
[1] 411
naran@Naran:~/praktikum1$ [sudo] password for naran:
ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   2.9  0.3  21688 12748 ?        Ss   17:05   0:01 /sbin/init
root         2   0.0  0.0   3072  1920 ?        SL   17:05   0:00 /init
root         7   0.0  0.0   3072  1792 ?        SL   17:05   0:00 plan9 --c
root        52   0.8  0.4  50440 19212 ?        S<s  17:05   0:00 /usr/lib/
root       101   0.4  0.1  25204  6528 ?        Ss   17:05   0:00 /usr/lib/
systemd+   132   0.3  0.3  21456 12672 ?        Ss   17:05   0:00 /usr/lib/
systemd+   136   0.1  0.1  91024  7680 ?        Ssl  17:05   0:00 /usr/lib/
root       161   0.0  0.0   4236  2688 ?        Ss   17:05   0:00 /usr/sbin
message+   162   0.1  0.1   9536  4864 ?        Ss   17:05   0:00 @dbus-dae
root       172   0.2  0.2  17964  8320 ?        Ss   17:05   0:00 /usr/lib/
root       174   0.3  0.3 1681852 12928 ?        Ssl  17:05   0:00 /usr/libe
root       176   0.0  0.0   3160  1920 hvc0    Ss+  17:05   0:00 /sbin/age
syslog     187   0.2  0.1  222508  5376 ?        Ssl  17:05   0:00 /usr/sbin
root       190   0.0  0.0   3116  1920 tty1     Ss+  17:05   0:00 /sbin/age
root       195   0.5  0.5 107032 22524 ?        Ssl  17:05   0:00 /usr/bin/
root       292   0.0  0.0   3076   896 ?        Ss   17:05   0:00 /init
root       293   0.0  0.0   3092  1152 ?        S    17:05   0:00 /init
naran     294   0.1  0.1   6072  5120 pts/0    Ss   17:05   0:00 -bash
root     295   0.0  0.1   6820  4480 pts/1    Ss   17:05   0:00 /bin/logi
naran     345   0.5  0.2  20308 11264 ?        Ss   17:05   0:00 /usr/lib/
naran     346   0.0  0.0  21152  3520 ?        S    17:05   0:00 (sd-pam)
naran     366   0.1  0.1   6072  5120 pts/1    S+   17:05   0:00 -bash
root     411   0.2  0.1  13624  6272 pts/0    T    17:06   0:00 sudo ./my
naran     412 50.0  0.1   8280  4096 pts/0    R+   17:06   0:00 ps aux

[1]+  Stopped                  sudo ./my_daemon

```

```

C:\Users\naran>wsl
naran@Naran:/mnt/c/Users/naran$ nano my_daemon.c
naran@Naran:/mnt/c/Users/naran$ gcc -o my_daemon my_daemon.c
naran@Naran:/mnt/c/Users/naran$ sudo ./my_daemon &
[1] 445
naran@Naran:/mnt/c/Users/naran$ [sudo] password for naran:
Naran41
Naran41: command not found

[1]+  Stopped                  sudo ./my_daemon
naran@Naran:/mnt/c/Users/naran$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.9  0.3  21656 12652 ?        Ss   16:19   0:03 /sbin/init
root         2  0.0  0.0   3072  1792 ?        Sl   16:19   0:00 /init
root         7  0.0  0.0   3072  1792 ?        Sl   16:19   0:00 plan9 --c
root        58  0.2  0.3  66744 15640 ?        S<s  16:19   0:00 /usr/lib/
root       108  0.0  0.1  25228  6656 ?        Ss   16:19   0:00 /usr/lib/
systemd+  123  0.0  0.3  21456 12672 ?        Ss   16:19   0:00 /usr/lib/
systemd+  124  0.0  0.1  91024  7808 ?        Ssl  16:19   0:00 /usr/lib/
root       170  0.0  0.0   4236  2560 ?        Ss   16:19   0:00 /usr/sbin
message+  171  0.0  0.1   9636  5248 ?        Ss   16:19   0:00 @dbus-dae
root       178  0.0  0.2  17968  8448 ?        Ss   16:19   0:00 /usr/lib/
root       180  0.0  0.3 1681852 12416 ?        Ssl  16:19   0:00 /usr/libe
syslog    189  0.0  0.1  222508  5760 ?        Ssl  16:19   0:00 /usr/sbin
root       196  0.0  0.0   3160  1920 hvc0     Ss+  16:19   0:00 /sbin/age
root       199  0.0  0.0   3116  1920 tty1     Ss+  16:19   0:00 /sbin/age
root       206  0.0  0.5 107032 22784 ?        Ssl  16:19   0:00 /usr/bin/
root       304  0.0  0.0   3076   896 ?        Ss   16:19   0:00 /init
root       305  0.0  0.0   3092  1152 ?        S    16:19   0:00 /init
naran     306  0.0  0.1   6072  5120 pts/0    Ss   16:19   0:00 -bash
root       307  0.0  0.1   6824  4608 pts/1    Ss   16:19   0:00 /bin/logi
naran     357  0.1  0.2  20308 11136 ?        Ss   16:19   0:00 /usr/lib/
naran     358  0.0  0.0   21148  3516 ?        S    16:19   0:00 (sd-pam)
naran     384  0.0  0.1   6072  4992 pts/1    S+   16:19   0:00 -bash
root       445  0.0  0.1  13624  6400 pts/0    T    16:24   0:00 sudo ./my
root       455  0.0  0.0   25232  3604 ?        S    16:25   0:00 (udev-wor
naran     456 33.3  0.1   8280  4224 pts/0    R+   16:25   0:00 ps aux
naran@Naran:/mnt/c/Users/naran$ sudo journalctl -t my_daemon -n 10

```

=== Information Maintenance System Calls ===

```

Process ID: 817
Parent Process ID: 294
User ID: 1000
Group ID: 1000
Current time: Wed Oct 15 11:38:18 2025
Clock ticks since boot: 1718434434
Sleeping for 2 seconds...
Awake!

```

```

mysnctl> exit
naran@Naran:/mnt/c/Users/naran$ nano process_syscall.c
naran@Naran:/mnt/c/Users/naran$ gcc -o process_syscall prosesess_syscall.c
cc1: fatal error: prosesess_syscall.c: No such file or directory
compilation terminated.
naran@Naran:/mnt/c/Users/naran$ gcc -o process_syscall process_syscall.c
naran@Naran:/mnt/c/Users/naran$ ./process_syscalls
-bash: ./process_syscalls: No such file or directory
naran@Naran:/mnt/c/Users/naran$ ./process_syscall
=== Process Control System Calls ===
Parent PID: 720
Parent waiting for child (PID: 721)...
Child PID: 721
Child's Parent PID: 720
Child executing 'ls' command...
total 30248
drwxrwxrwx 1 naran naran    4096 Dec 17  2024 AppData
lrwxrwxrwx 1 naran naran     34 Dec 16  2024 'Application Data' -> /mnt/c/

```

```
myshell> ls
AppData
'Application Data'
Contacts
Cookies
CrossDevice
Documents
Downloads
Favorites
IntelGraphicsProfiles
Links
'Local Settings'
Music
'My Documents'
NTUSER.DAT
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TM.blf
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000001.regtra
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000002.regtra
NetHood
OneDrive
PrintHood
Recent
'Saved Games'
Searches
SendTo
'Start Menu'
Templates
Videos
mudah
mudah.c
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini
myshell> pwd
/mnt/c/Users/naran
myshell> date
Wed Oct 15 10:52:40 WIB 2025
myshell> echo hello, naran
hello, naran
myshell> |
```